SIMOCODE pro C to SEL RTAC Monitoring System

Commissioning, Mapping, and Troubleshooting Guide

ProSoft PLX51-PBM PROFIBUS DP to Modbus TCP Gateway

System / Area	Motor monitoring system using SIMOCODE pro C, PLX51-PBM, and SEL RTAC 3555
Document Type	Handover, commissioning, and troubleshooting process
Current Scope	Two SIMOCODE pro C devices. Future devices can be added using the same pattern.
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Contents

Table of Revisions

Revision	Date	Description	Author	Reviewer	Approved
0	2026/06/19	Initial draft for two-device commissioning and troubleshooting process	Naman Arora		
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1. Purpose and Scope

This document explains how to commission, map, test, and troubleshoot the communication path between Siemens SIMOCODE pro C devices, a ProSoft PLX51-PBM gateway, and an SEL RTAC 3555.

The document is written for engineers and technologists who may need to troubleshoot the system later or add more SIMOCODE devices after the initial two-device setup.

1.1 What this document covers

- How to check or add a SIMOCODE pro C in TIA Portal / SIMOCODE ES.
- How to set the SIMOCODE PROFIBUS address and basic motor protection settings.
- How to configure the PLX51-PBM as a PROFIBUS DP master and Modbus TCP server for the RTAC.
- How to map SIMOCODE PROFIBUS data into Modbus TCP registers.
- How to connect the PROFIBUS daisy chain using Siemens DB9 PROFIBUS connectors.
- How to configure the RTAC as a Modbus TCP client and verify tags.
- How to troubleshoot common faults such as BF red, BUS LED off/blinking, general fault, wrong register values, and Modbus timeouts.

1.2 Current scope

Item	Current document scope
SIMOCODE devices	Two devices only. Future devices will be added using the same address and register pattern.
HMI	High-level HMI structure is included. Final Diagram Builder screenshots will be added later.
Plant configuration files	Final plant files are not yet included. Update this document after the final files are built.
Screenshots	Placeholders are included. Replace each placeholder with actual site screenshots.

Plain-language rule: Before working on the HMI, prove the raw data first. The chain must be green and data must update in RTAC tags before building screen graphics.

2. System Overview

The system converts motor protection and status data from SIMOCODE pro C into RTAC/HMI tags.

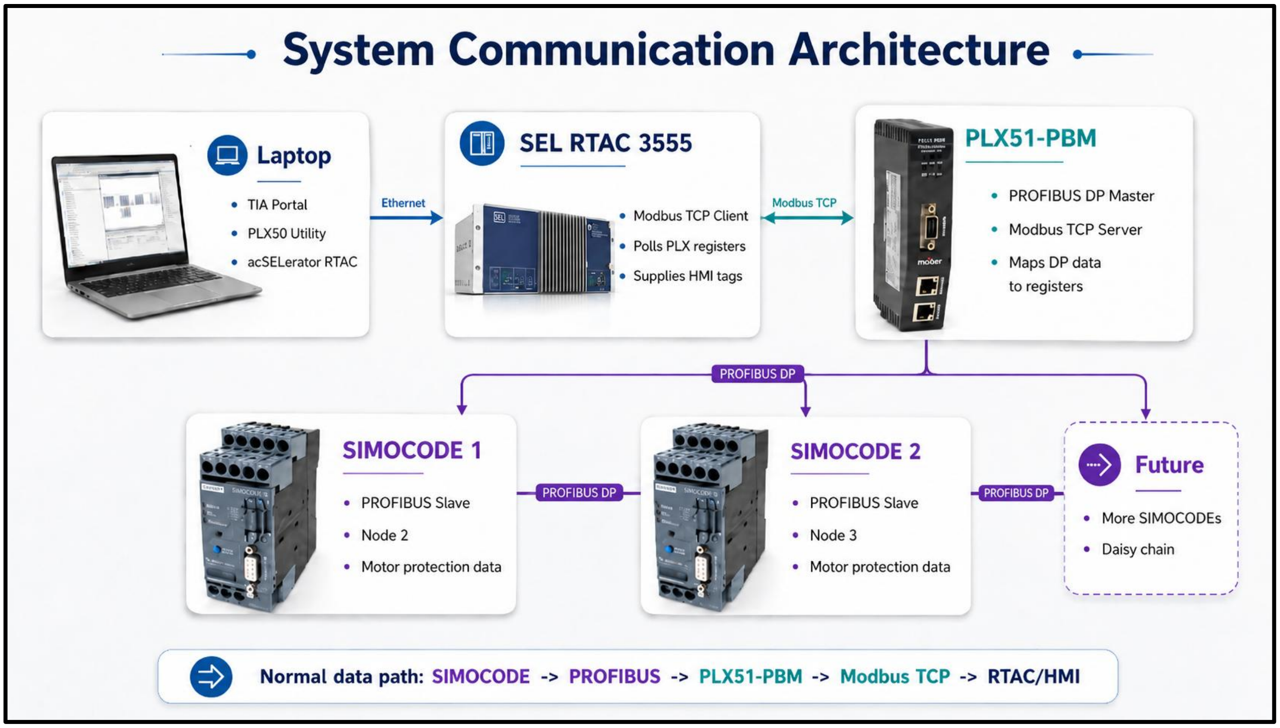

Figure 1 - Normal data path from SIMOCODE to RTAC

2.1 Device roles

Device	Role	Important settings
SIMOCODE pro C	Motor management and protection devices. It is a PROFIBUS DP slave.	Unique PROFIBUS address, correct current measuring module, correct motor protection settings, correct cyclic send data.
PLX51-PBM	Gateway. It is the PROFIBUS DP master and exposes mapped data to Ethernet.	Mode = Standalone Master. Primary Interface = Modbus TCP/IP Slave when the RTAC is polling the PLX.
SEL RTAC 3555	Modbus TCP client. It polls the PLX and creates tags for HMI and logic.	Same IP subnet as PLX. Modbus server IP = PLX IP. Polls must match PLX Modbus offsets.
Ethernet switch	Connect laptop, PLX, and RTAC on the same network.	Correct DC power and Ethernet links active.
Laptop	Used for TIA Portal, PLX50 Configuration Utility, ACSELERATOR RTAC, and Diagram Builder.	Wired Ethernet on same subnet. Wi-Fi can be disabled during bench setup to avoid network confusion.

2.2 Current two-device example

Device	PROFIBUS role	PROFIBUS address	Suggested register base	Notes
PLX51-PBM	DP Master	1	Not applicable	Master address must not duplicate any SIMOCODE address.
SIMOCODE 1	DP Slave	2	0	First test device.
SIMOCODE 2	DP Slave	3	10	Second test device.

Address rule: Every PROFIBUS station must have a unique address. Do not use the same address for the PLX and a SIMOCODE.

3. Important Rules Before Starting

- **Work safely.** Follow site electrical safety rules, lockout/tagout rules, and arc-flash procedures. Only qualified personnel should work inside energized MCC or control cabinets.
- **Back up before changing.** Upload or save the existing SIMOCODE, PLX, and RTAC projects before editing.
- **Keep one change at a time.** Change one setting, download, then test. This makes troubleshooting easier.
- **Match software to hardware.** The current measuring module in the TIA project must match the physical module connected to the SIMOCODE.
- **Check the ribbon/system cable.** The SIMOCODE basic unit must be connected to the current measuring module using the correct Siemens system connection cable.
- **Use the correct upstream protocol.** For SEL RTAC polling the PLX, configure the PLX Ethernet side as Modbus TCP/IP Slave/Server. Use EtherNet/IP only for a Rockwell/Logix system.
- **Do not show unavailable values.** With SIMOCODE pro C and a current measuring module, use fault, warning, overload prewarning, and current/load percent. Do not show voltage, kW, kWh, or power factor unless the hardware supports those measurements.
- **Verify raw registers first.** Do not build or troubleshoot HMI graphics until the RTAC raw Modbus registers are updated.

4. Tools, Files, and Information Needed

Item	Purpose	Notes
TIA Portal / SIMOCODE ES	Configure and diagnose SIMOCODE pro C	Used for offline configuration and online parameter checks.
Siemens PC cable 3UF7940-0AA00-0	Connect laptop to SIMOCODE system interface	Used when checking or changing SIMOCODE parameters directly.
Correct SIMOCODE GSD file	Allows PLX to identify and configure SIMOCODE	Use the correct GSD for SIMOCODE pro C version used on site.
PLX50 Configuration Utility	Configure PLX51-PBM and PROFIBUS-to-Modbus map	Used for device discovery, GSD import, slot configuration, and diagnostics.
SEL ACSELERATOR RTAC	Configure RTAC Modbus client and tags	Use project firmware/version matching the installed RTAC.
SEL Diagram Builder	Create HMI screens	Final HMI section will be expanded after tags are proven.
PROFIBUS DP cable and Siemens DB9 connectors	Connect PLX and SIMOCODE devices	Use proper PROFIBUS DP cable for permanent installation.
Ethernet switch	Connect PLX, RTAC, and laptop	Bench switch can be 24 VDC industrial Ethernet switch.

5. SIMOCODE pro C Configuration in TIA Portal / SIMOCODE ES

Use this section when adding a new SIMOCODE, checking a SIMOCODE, or fixing a SIMOCODE general fault/configuration fault.

5.1 Connect to SIMOCODE

1. Connect the Siemens PC cable 3UF7940-0AA00-0 from the laptop to the SIMOCODE system interface.
2. Open TIA Portal / SIMOCODE ES.
3. Open the existing project or create a new project if this is a new device.
4. Upload and save the existing online parameter set before making changes, if the device has already been used.

Figure 2 - Laptop cable connected to SIMOCODE pro C system interface

5.2 Offline configuration checks

5. Add the correct SIMOCODE device type. For this project use SIMOCODE pro C V2.0 if that matches the installed hardware.
6. Open Device Configuration.
7. Check the current measuring module in the project.
8. Compare the project module with the physical current measuring module installed beside the SIMOCODE.
9. If the module is wrong, right-click the current measuring module and use Change Device to select the correct module.
10. Compile the project. Use Rebuild All if needed.

Common general fault cause: A mismatch between the configured current measuring module and the installed module can create a configuration fault. A missing or loose system/ribbon cable can also create a general fault.

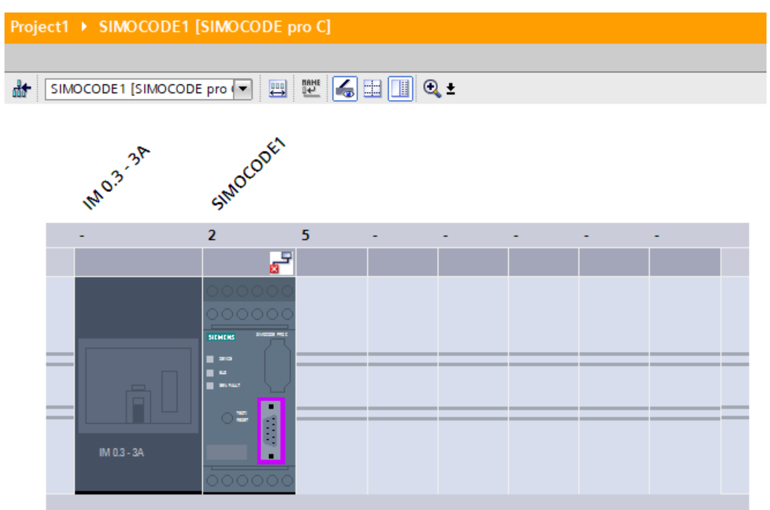

Figure 3 - TIA device configuration showing SIMOCODE pro C and current measuring module

5.3 Online parameter checks

11. Go Online with the SIMOCODE.
12. Open Parameters.

13. Open Fieldbus Interface and check the PROFIBUS node address.
14. Set the node address to an unused PROFIBUS address. For this two-device example, SIMOCODE 1 = node 2 and SIMOCODE 2 = node 3.
15. Check the PROFIBUS baud rate / transmission speed and highest station address. These must match the PLX PROFIBUS network settings.
16. Open Configuration and select the required control function. For this project the starting point is Overload Relay / monitoring, not remote motor control.
17. Open Motor Protection and set Is1 to the correct motor full-load current or test current. This affects the Imax % of Is value used on the HMI.
18. Open Outputs > Cyclic Send Data and check the data mapped from SIMOCODE to the PROFIBUS master.
19. Download parameter changes to the device.
20. Upload/read back from the device to confirm that the online device matches the project.
21. Open Commissioning / Diagnostics and check active faults, warnings, and status information.
22. Save the project, go offline, and close TIA Portal when finished.

Parameter	What to check	Why it matters
PROFIBUS address	Unique node address for each SIMOCODE	Duplicate addresses stop communication.
Baud rate / speed	Same as the PLX PROFIBUS network	Mismatch prevents data exchange.
Highest station address	Same range as the PLX network	Incorrect range can block discovery or polling.
Current measuring module	Project hardware matches physical hardware	Mismatch can cause general fault/configuration error.
System cable	Basic unit connected to current module	Missing/loose cable can cause general fault.
Is1 current	Correct motor/test current	This scales Imax % of Is.
Cyclic Send Data	Status word and current/load value are available	These values are mapped to Modbus for RTAC/HMI.

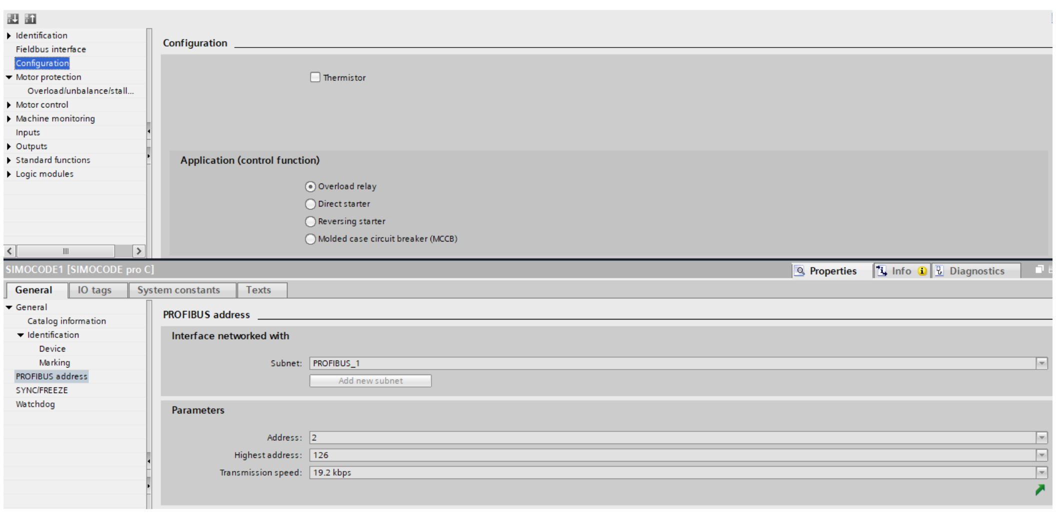

Figure 4 - Fieldbus Interface page showing PROFIBUS address

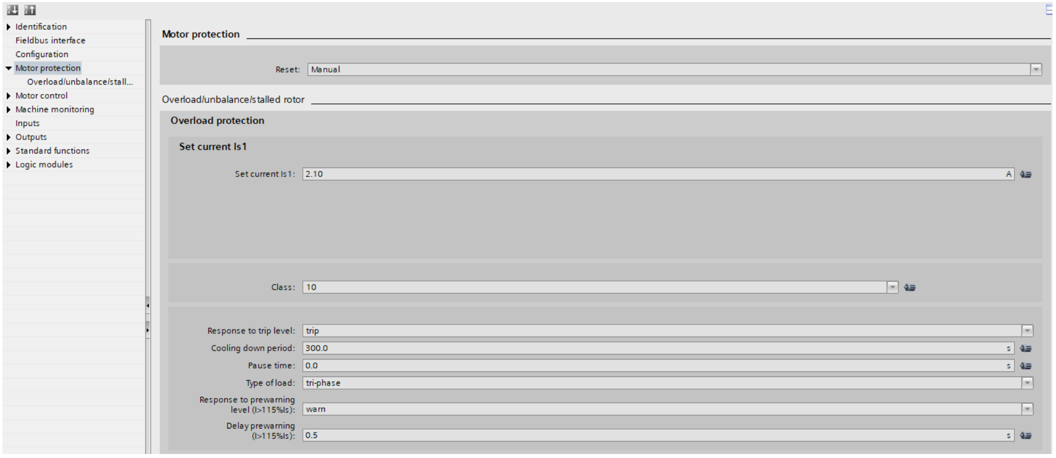

Figure 5 - Motor Protection page showing Is1 current

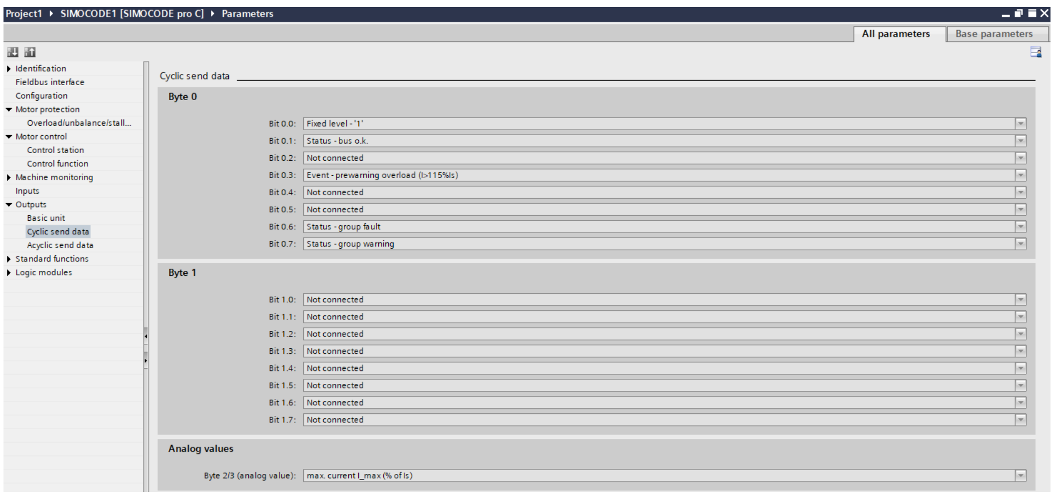

Figure 6 - Cyclic Send Data page showing mapped status/current data

5.4 SIMOCODE LED checks

LED / condition	Good sign	If not good, check
DEVICE LED	Green	SIMOCODE power, device hardware, active fault.
BUS LED	Green when communicating with the PROFIBUS master	PROFIBUS cable, node address, baud rate, PLX config, termination.
GEN. FAULT	Off	Current module mismatch, missing system cable, parameter mismatch, motor protection fault, device diagnostics.

6. PLX51-PBM Gateway Configuration

The PLX51-PBM is the bridge between PROFIBUS DP and Ethernet. In this project it acts as the PROFIBUS DP master. The RTAC polls the PLX over Modbus TCP.

> **Important correction:** For this RTAC system, the PLX Ethernet side should be configured for Modbus TCP/IP Slave/Server so the RTAC can poll it. Do not select EtherNet/IP unless the upstream controller is Rockwell/Logix.

6.1 Set or recover the PLX IP address

23. Connect 24 VDC power wiring to the PLX power terminals, but keep power off until ready.
24. Connect the laptop Ethernet cable to the PLX or to the same Ethernet switch as the PLX.
25. Open Command Prompt on the laptop and run ipconfig. Record the laptop IPv4 address and subnet mask.
26. If the PLX IP is unknown, power up the PLX with DIP switch 2 ON to force DHCP mode.
27. Open PLX50 Configuration Utility > Tools > DHCP Server.
28. Assign the PLX an IP address in the same subnet as the laptop. Example: if the laptop is 192.168.1.10, use PLX 192.168.1.11.
29. After the IP address is assigned and confirmed, set DIP switch 2 back OFF. Do not leave DIP 2 ON for normal operation.
30. Use Target Browser or Ping to confirm the laptop can reach the PLX.

DIP switch	Use	Normal position
DIP 1	Safe Mode recovery	OFF
DIP 2	Forces DHCP mode when powered up	OFF after IP setup
DIP 3	Locks configuration from being overwritten	OFF while commissioning
DIP 4	Forces IP 192.168.1.100 at boot	OFF unless recovering IP

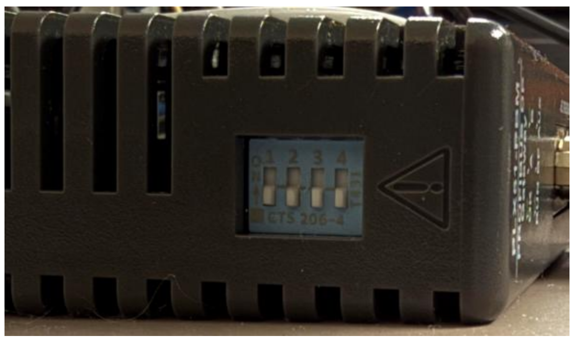

Figure 7 – PLX51 - PBM DIP switches

6.2 Create the PLX project

31. Open PLX50 Configuration Utility.
32. Click File > New.
33. Click Device > Add.
34. Select PLX51-PBM PROFIBUS Gateway Master/Slave module.
35. Open the PLX configuration window.
36. Set the instance name, description, and IP address.

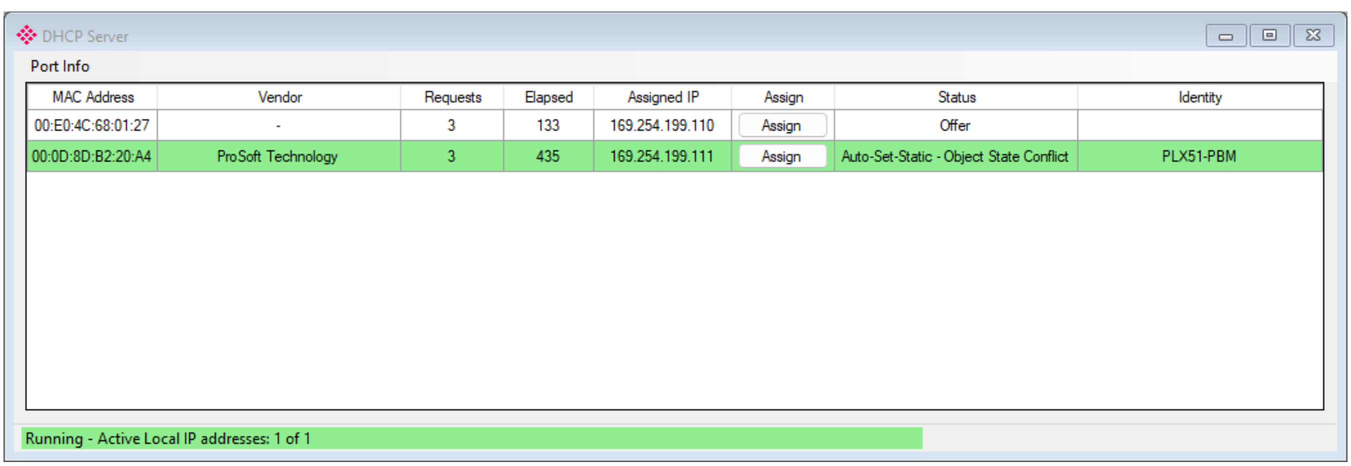

Figure 8 – Add PLX51 - PBM in PLX50 Configuration Utility

6.3 PLX general settings

Setting	Use for this project	Notes
Mode	Standalone Master	PLX is the PROFIBUS DP master.
Primary Interface	Modbus TCP/IP Slave	RTAC is the Modbus TCP client/master that polls the PLX.
IP Address	Project/site PLX IP	Must be in the same network as RTAC.
Local Node Number	1	Use a consistent Modbus node number. Match RTAC if node is used.
Modbus TCP Port	502	Default Modbus TCP port.
PROFIBUS Station Address	1	PLX master address.
Highest Station Address	126	Must allow all SIMOCODE addresses used.
Baud Rate	Match TIA/SIMOCODE	All PROFIBUS devices must use compatible speed.
Modbus Comms Fail	Force to Clear	Prevents PROFIBUS from being forced offline during setup when RTAC is not yet polling.

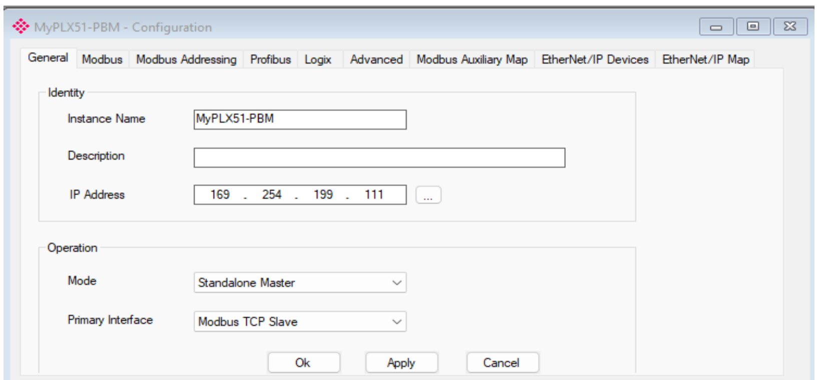

Figure 9 – PLX General Tab

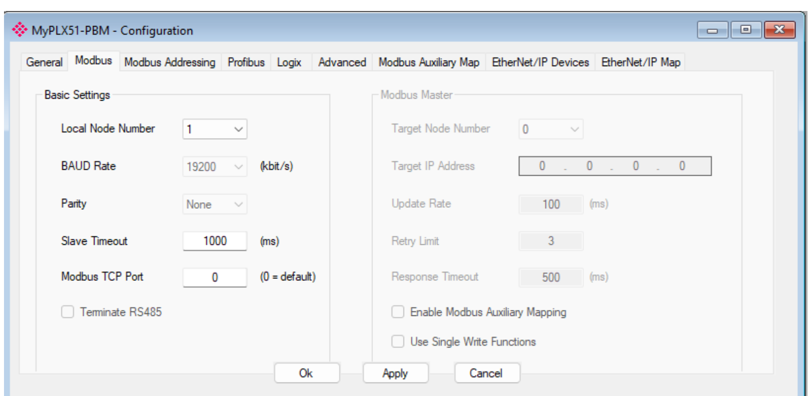

Figure 10 – PLX Modbus Tab

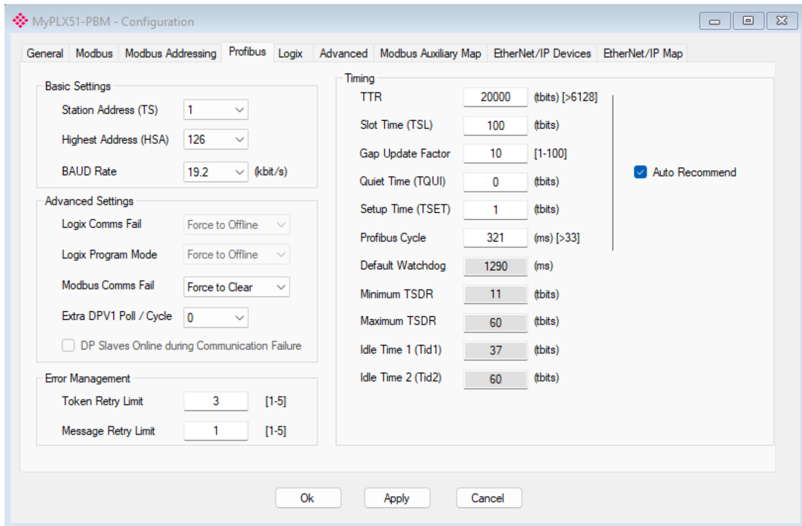

Figure 11 – PLX Profibus Tab

6.4 Add SIMOCODE GSD and devices

37. Download or locate the correct GSD file for the SIMOCODE pro C version used on site.
38. In PLX50, open GSD File Management and add the SIMOCODE GSD file if it is not already in the catalog.
39. Right-click PROFIBUS Devices and select Add PROFIBUS Device.
40. Choose the SIMOCODE pro C device from the GSD catalog.
41. Set the instance name. Example: SIMOCODE_1 or RELAY_1.
42. Open the PROFIBUS Configuration tab and set the node address to match the SIMOCODE online address in TIA.
43. Repeat for the second SIMOCODE.
44. Enable DPV1 if diagnostic/acyclic access is required by the setup.

Device	PLX instance name	PROFIBUS address	Action
SIMOCODE 1	SIMOCODE_1	2	Add from GSD and set node 2.
SIMOCODE 2	SIMOCODE_2	3	Add from GSD and set node 3.

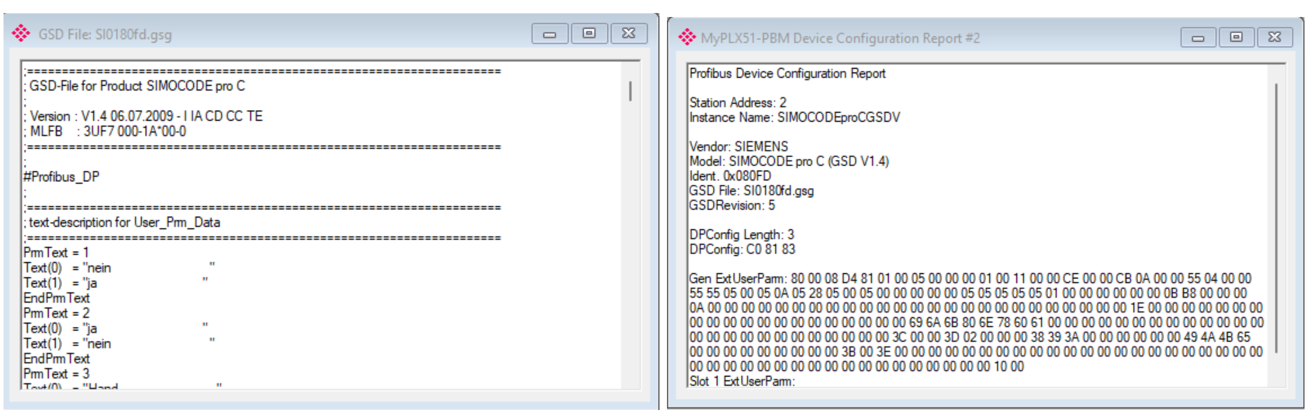

Figure 12 – SIMOCODE Pro C GSD File & Configuration Information

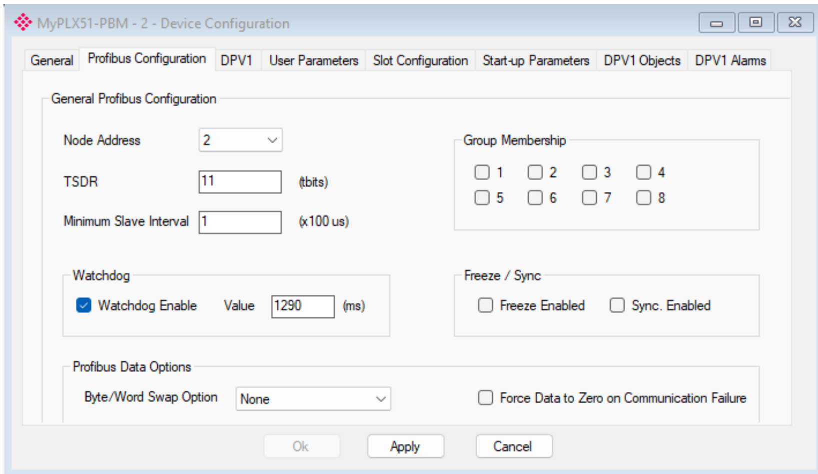

Figure 13 – PROFIBUS Configuration in PLX

6.5 Slot configuration and mapping

This is where the PROFIBUS data is mapped to Modbus registers.

Term	Simple meaning
PROFIBUS DP Offset	The byte position inside the SIMOCODE cyclic data telegram.
Modbus Offset	The register location inside the PLX that the RTAC will read or write.
Byte length	How many bytes are moved. One Modbus register is 2 bytes.
Status word	A 2-byte word containing status bits such as warning/fault/prewarning.
Control word	A 2-byte word that can be used later for commands if control is approved and mapped.
Imax % of Is	Current-based load value. It is not kW.

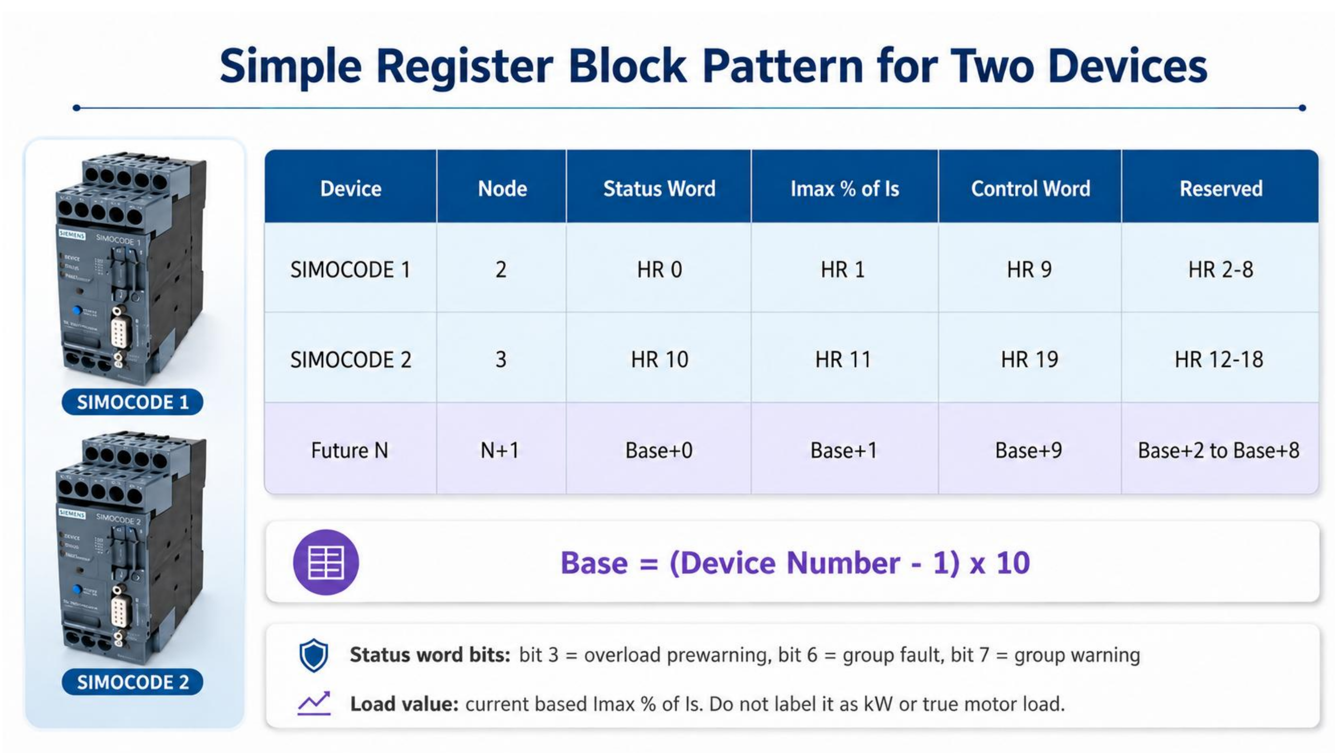

Figure 14 - Suggested register block pattern for two SIMOCODE devices

6.6 Suggested two-device PLX register map

Device	Node	Signal	Direction	PLX Modbus offset	RTAC use
SIMOCODE 1	2	Status Word	SIMOCODE -> RTAC	0	Break into fault/warning/prewarning bits.
SIMOCODE 1	2	Imax % of Is	SIMOCODE -> RTAC	1	Display as Load % / Imax % of Is.
SIMOCODE 1	2	Control Word	RTAC -> SIMOCODE	9	Reserved unless control is approved.
SIMOCODE 2	3	Status Word	SIMOCODE -> RTAC	10	Break into fault/warning/prewarning bits.
SIMOCODE 2	3	Imax % of Is	SIMOCODE -> RTAC	11	Display as Load % / Imax % of Is.
SIMOCODE 2	3	Control Word	RTAC -> SIMOCODE	19	Reserved unless control is approved.

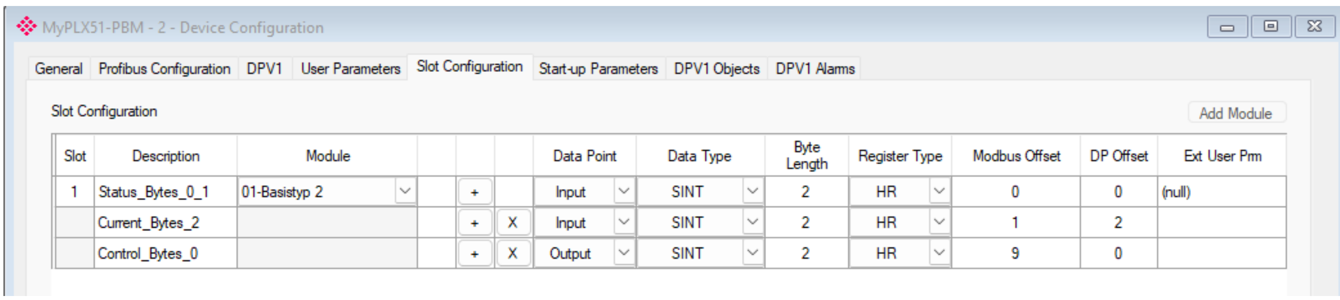

Figure 15 – PROFIBUS to MODBUS Mapping

Register address convention: Some software shows zero-based offsets, while other software shows 40001-based Modbus addresses. Confirm the convention before troubleshooting. Example: PLX offset 0 may appear as 40001 in a Modbus client.

6.7 Download and check PLX status

45. Click Apply after each major change.
46. Download the project to the PLX.
47. Open PLX Status / Online Status.
48. Check that the configuration is valid.
49. Check that the PLX mode is Standalone Master.
50. Open PROFIBUS live list / status.
51. Confirm node 1 master is live and each SIMOCODE slave is online.
52. Open each SIMOCODE device status and confirm Data Exchange Active.
53. Open Modbus Statistics and confirm Tx/Rx counts increase when RTAC is polling.
54. Open Event Log and save screenshots before clearing messages.

Good status	Meaning
Config Valid	The PLX project was accepted.
PROFIBUS State = Clear or Operate	The master is not offline. Operate is preferred for final operation.
SIMOCODE Online	The PLX can see the slave.
Data Exchange Active	The PLX is exchanging DPV0 process data with the slave.
IO bytes/sec not zero	Cyclic PROFIBUS data is moving.
Modbus Tx/Rx increasing	RTAC or another Modbus client is polling the PLX.

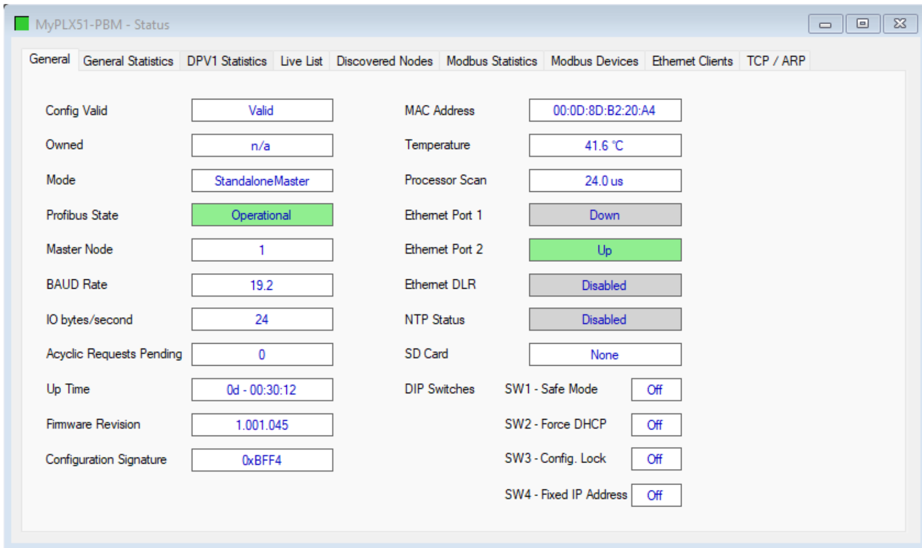

Figure 16 – General Status Information after Mapping

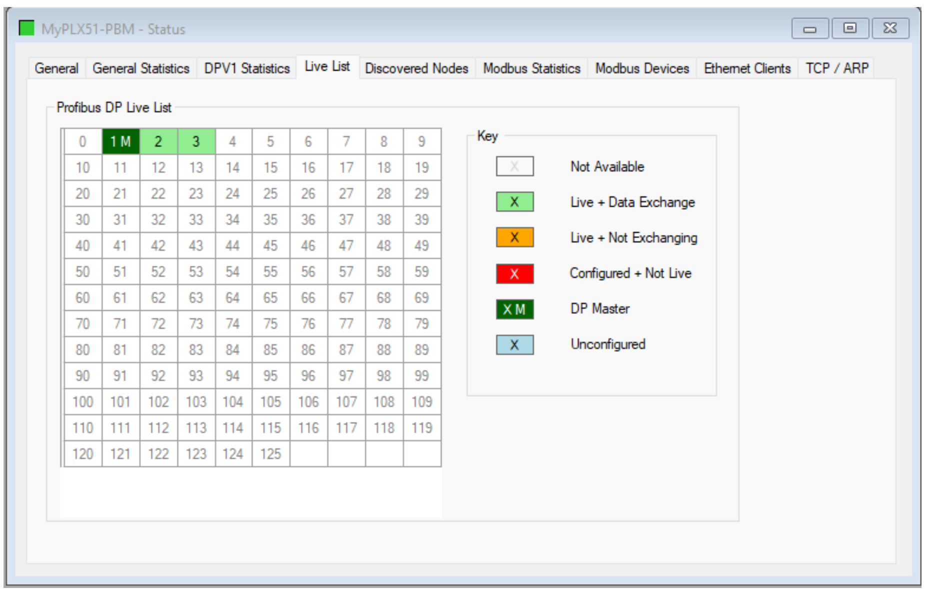

Figure 17 - PLX device status showing Data Exchange Active & Live list of Devices

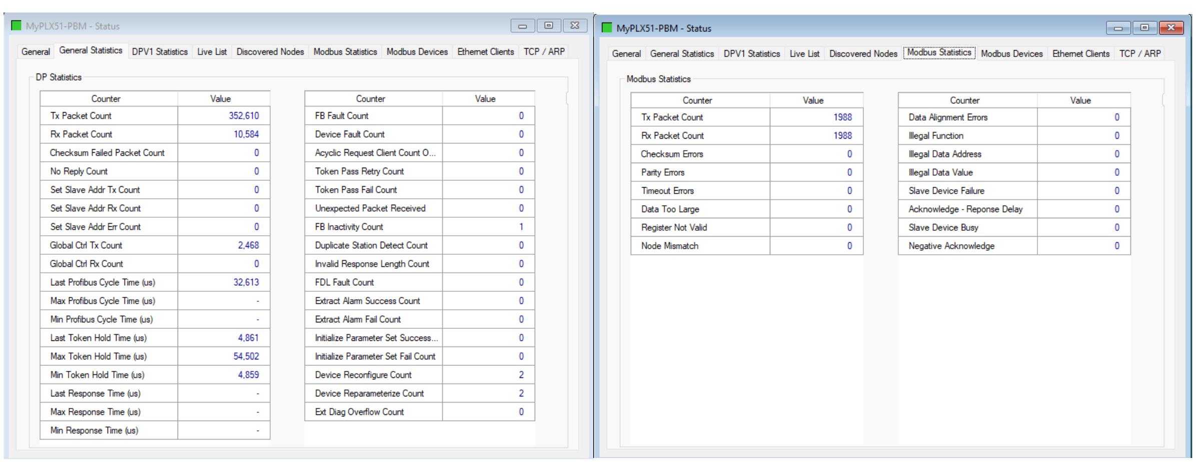

Figure 18 - PLX General & Modbus statistics showing Tx/Rx counts increasing

7. PROFIBUS Cable, DB9 Connector, and Daisy Chain Rules

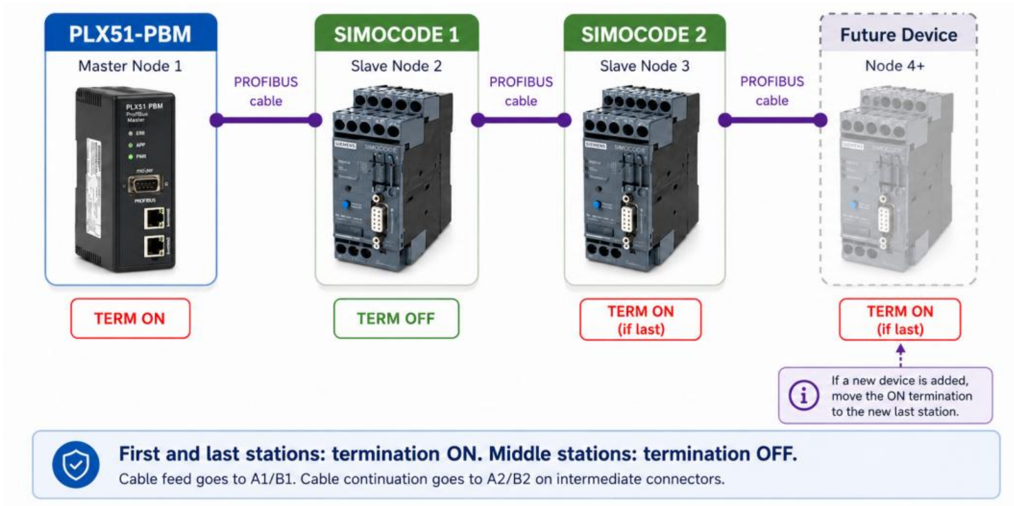

Figure 19 - PROFIBUS daisy chain and termination rule

7.1 Connector wiring rules

- Use proper PROFIBUS DP cable for permanent installation.
- Keep the same conductor on A and the same conductor on B through the full chain.
- For Siemens connectors, the manual recommendation is A = green and B = red.
- For the first and last stations, connect the cable on A1/B1 and turn termination ON.
- For middle stations, feed in on A1/B1, continue out on A2/B2, and turn termination OFF.
- The cable shield must contact the connector shield/contact element. Do not leave the shield floating inside the connector.

7.2 Normal LED result after PROFIBUS wiring

Device	Expected LED result	If not correct
PLX51-PBM	OK green, RUN green, BF off	Check PROFIBUS wiring, node addresses, baud rate, GSD, slot config, and termination.
SIMOCODE pro C	DEVICE green, BUS green, GEN. FAULT off	If BUS is off/blinking, check PROFIBUS. If GEN. FAULT is red, check SIMOCODE diagnostics and hardware configuration.

Cable warning: The orange Belden 3076F fieldbus cable may work in a low-speed bench test, but it is not the preferred permanent PROFIBUS DP cable. Use proper PROFIBUS DP Type A cable for plant wiring.

Figure 20 - Actual PROFIBUS DB9 connector termination switch positions

Figure 21 - PLX and SIMOCODE LEDs after valid data exchange

8. SEL RTAC 3555 Modbus TCP Configuration

After PROFIBUS communication is valid, connect the RTAC. The RTAC should poll the PLX over Modbus TCP.

8.1 Ethernet layout

55. Power the Ethernet switch using the correct DC supply and pinout.
56. Connect PLX Ethernet to the switch.
57. Connect laptop Ethernet to the switch.
58. Connect RTAC Ethernet to the switch.
59. Set laptop, PLX, and RTAC IP addresses in the same subnet.
60. Ping each device before opening ACSELERATOR RTAC.

Device	Example IP	Notes
Laptop	192.168.1.10	Example only. Use actual site network.
PLX51-PBM	192.168.1.11	RTAC Modbus server IP points to this address.
SEL RTAC 3555	192.168.1.12	Same subnet as laptop and PLX.
Gateway	Blank or site gateway	For bench testing, gateway may not be needed.

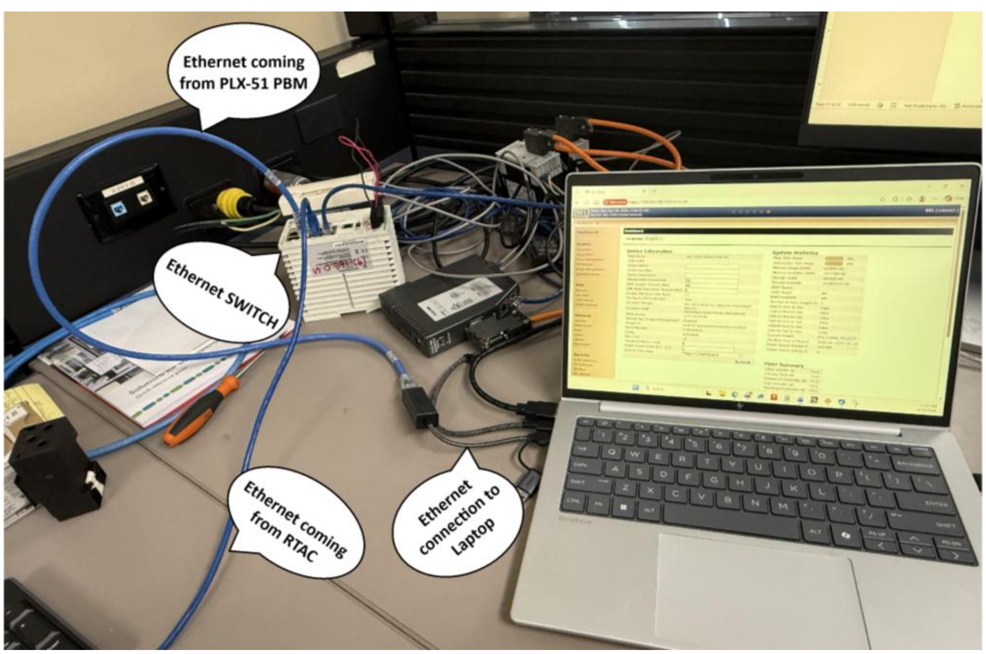

Figure 22 - Ethernet switch connection between laptop, PLX, and RTAC

8.2 Create RTAC Modbus client

61. Open SEL ACSELERATOR RTAC.
62. Create a new project or open the existing RTAC project for the correct RTAC firmware/version.
63. Insert the SEL RTAC 3555 device if it is not already in the project.
64. Add a Modbus protocol connection.
65. Select Client - Ethernet, because the RTAC will poll the PLX.
66. In the Modbus protocol settings, set Server IP Address to the PLX IP address.
67. Set the Modbus TCP port to 502 unless the site uses a different port.
68. Set the poll period. Start with a moderate polling period during commissioning, then adjust if needed.

Password rule: Do not write RTAC passwords in this document. Use approved site credential storage and access controls.

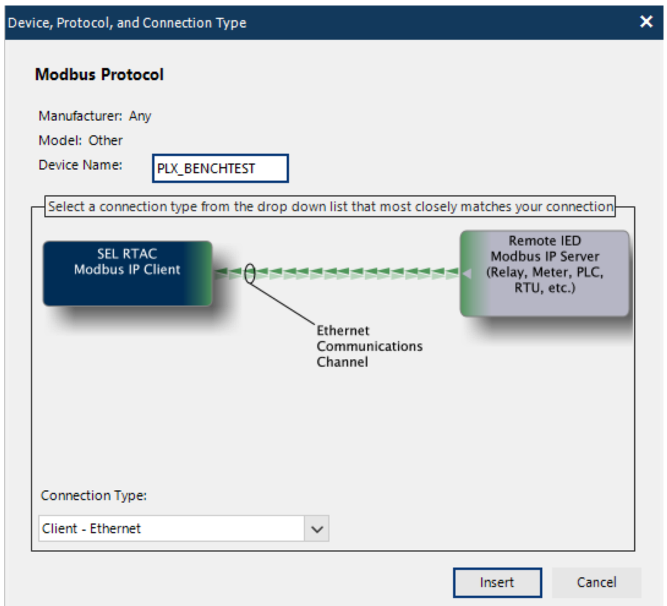

Figure 23 - RTAC Modbus Client - Ethernet connection settings

8.3 Add raw holding registers

Start by reading raw words from the PLX. Only break words into bits after the raw word changes correctly.

RTAC tag	PLX register	Type	Purpose
SC1_STATUS_WORD	HR 0	16-bit word	SIMOCODE 1 status word.
SC1_IMAX_PERCENT	HR 1	16-bit word	SIMOCODE 1 Imax % of Is.
SC1_CONTROL_WORD	HR 9	16-bit word	Reserved control word.
SC2_STATUS_WORD	HR 10	16-bit word	SIMOCODE 2 status word.
SC2_IMAX_PERCENT	HR 11	16-bit word	SIMOCODE 2 Imax % of Is.
SC2_CONTROL_WORD	HR 19	16-bit word	Reserved control word.

Enable	Tag Name	Tag Type	Tag Alias	Register Address Start	Register Address Stop	Variation
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00000	INC	SIMOCODE_STATUS_WORD	0	0	16 bit signed MSB
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00000.status	INS	SIMOCODE_STATUS_WORD.status	0	0	
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00000.oper	operINC	SIMOCODE_STATUS_WORD.oper	0	0	
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00001	INC	SIMOCODE_IMAX_RAW	1	1	16 bit signed MSB
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00001.status	INS	SIMOCODE_IMAX_RAW.status	1	1	
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00001.oper	operINC	SIMOCODE_IMAX_RAW.oper	1	1	
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00002	INC	SIMOCODE_CONTROL_WORD	9	9	16 bit signed MSB
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00002.status	INS	SIMOCODE_CONTROL_WORD.status	9	9	
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00002.oper	operINC	SIMOCODE_CONTROL_WORD.oper	9	9	

Figure 24 - RTAC holding registers for status word, Load%, and control word

8.4 Break the status word into bits

The HMI needs simple Boolean tags. Break the status word into the required bits in the RTAC project.

Signal	Status word bit	HMI label	Alarm use
Overload prewarning	Bit 3	OL Prewarning	Medium priority alarm or warning.
Group fault	Bit 6	Fault	High priority alarm.
Group warning	Bit 7	Warning	Medium/low priority alarm.

Device	Boolean tag examples
SIMOCODE 1	SC1_OL_PREWARN, SC1_GROUP_FAULT, SC1_GROUP_WARNING
SIMOCODE 2	SC2_OL_PREWARN, SC2_GROUP_FAULT, SC2_GROUP_WARNING

Enable	Tag Name	Tag Type	Tag Alias	Register Address Start	Register Address Stop	Variation	Bit Reference
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00012	MDBC	SIMOCODE_OVERLOAD_EVENT	0	0		3
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00012.status	SPS	SIMOCODE_OVERLOAD_EVENT.status	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00012.operSet	operSPC	SIMOCODE_OVERLOAD_EVENT.operSet	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00012.operClear	operSPC	SIMOCODE_OVERLOAD_EVENT.operClear	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00013	MDBC	SIMOCODE_GENERAL_FAULT	0	0		6
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00013.status	SPS	SIMOCODE_GENERAL_FAULT.status	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00013.operSet	operSPC	SIMOCODE_GENERAL_FAULT.operSet	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00013.operClear	operSPC	SIMOCODE_GENERAL_FAULT.operClear	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00014	MDBC	SIMOCODE_GENERAL_WARNING	0	0		7
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00014.status	SPS	SIMOCODE_GENERAL_WARNING.status	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00014.operSet	operSPC	SIMOCODE_GENERAL_WARNING.operSet	0	0		
True	PLX51_PBM_SIMOCODE_MODBUS.HREG_00014.operClear	operSPC	SIMOCODE_GENERAL_WARNING.operC...	0	0		

Figure 25 – RTAC bit extraction tags from the status word

8.5 Add read polls and verify tags

69. Open Read Holding Register Polls.
70. Add a poll for the first register block. Example: start at HR 0 and read enough quantity to include HR 11 or HR 19 depending on the map.
71. Set the poll period.
72. Save the RTAC project.
73. Go online with the RTAC using its IP address.
74. Upload/download the project to the RTAC as required by site procedure.
75. Open the Tags view and expand the Modbus tags.
76. Confirm raw words update first.
77. Confirm Boolean bits change correctly when the SIMOCODE fault/warning conditions are forced or simulated.

Validation rule: If raw registers do not update in RTAC, do not troubleshoot the HMI. First check IP addresses,

RTAC poll configuration, PLX Modbus statistics, and PLX register offsets.

Figure 24 – RTAC read holding register polls

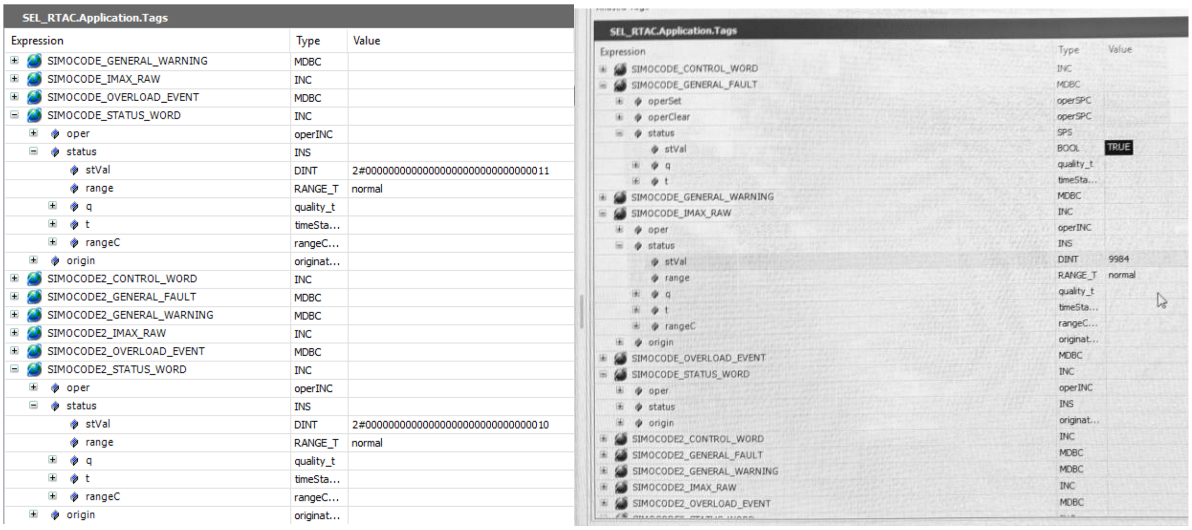

Figure 26 - RTAC online tags showing SIMOCODE data through PLX for STATUS_WORK & Load% (aka Imax%)

9. HMI / Diagram Builder Starting Point

The HMI should be a monitoring HMI first. Do not add start/stop control unless the control word is reviewed, approved, tested, and documented.

Version check: Before loading HMI pages, confirm the RTAC firmware, Diagram Builder software, and RTAC HMI runtime are compatible. This prevents a good HMI project from failing because the versions do not match.

9.1 First HMI screens

Screen	Purpose
01_MCC02502A1-P1	MCC A1 device list and status.
02_MCC02502A1-P2	MCC A2 device list and status.
03_MCC02502B1-P1	MCC B1 device list and status.
04_MCC02502B2-P2	MCC B2 device list and status.

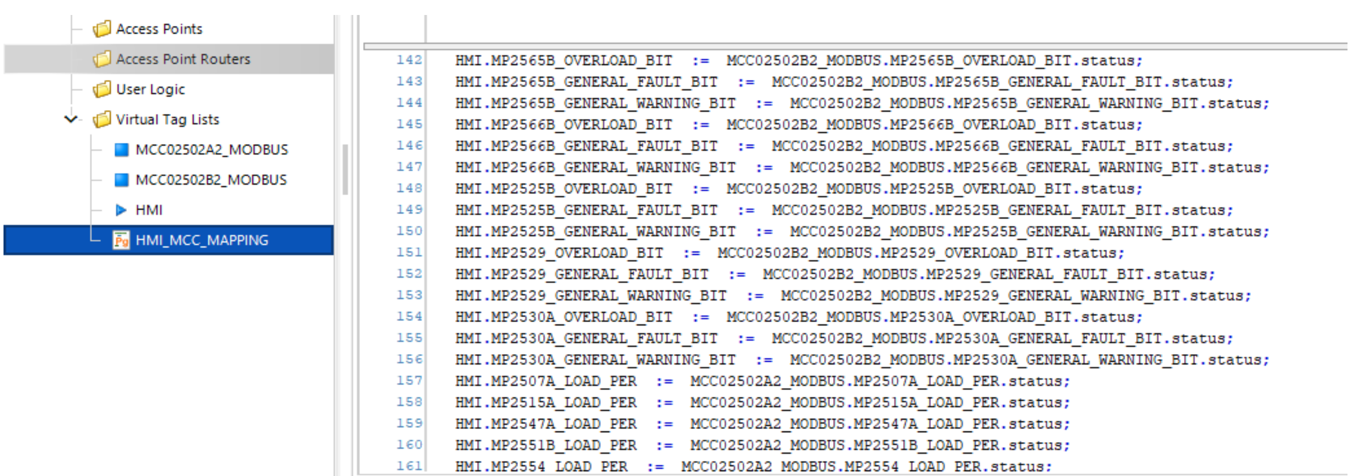

Figure 27 – HMI Status Bits Mapping

9.2 Values to show for each SIMOCODE

HMI value	RTAC tag source	Display wording
Overload Prewarning	Status word bit 3	OL Prewarning
Group Fault	Status word bit 6	Fault
Group Warning	Status word bit 7	Warning
Load / Imax % of Is	Imax percent word	Load % (Imax % of Is)
Communication Status	RTAC/PLX derived status	Comm OK / Comm Fail

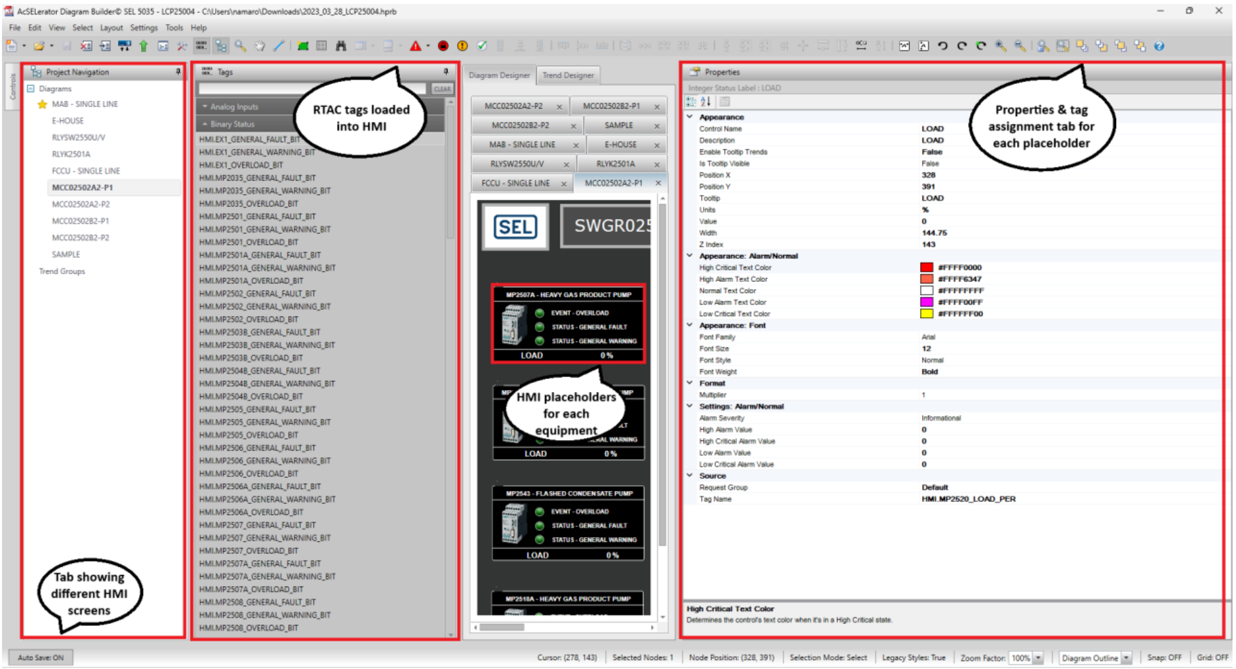

Figure 28 – HMI Builder Overview

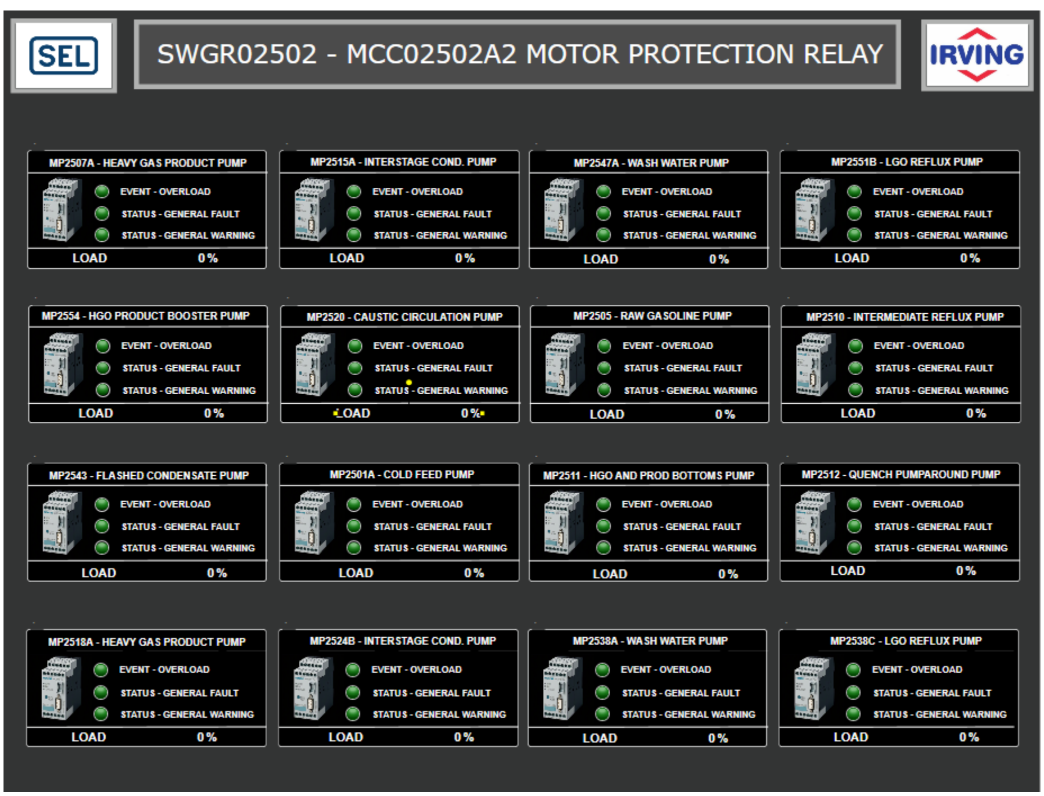

Figure 29 - MCC screen / motor list placeholder

10. Commissioning Test Plan

Test	Pass condition	Evidence to save
1. SIMOCODE hardware check	Correct module installed, system cable connected, no unexpected GEN. FAULT.	Photo of hardware and TIA device configuration.
2. SIMOCODE address check	Each SIMOCODE has a unique PROFIBUS address.	Screenshot of Fieldbus Interface address.
3. PLX project check	Config valid, Standalone Master, correct baud rate, correct HSA.	Screenshot of PLX general/status page.
4. PROFIBUS physical check	Termination ON only at first and last station. A/B and shield correct.	Photo of DB9 connectors and switches.
5. PLX live list	Each SIMOCODE online and Data Exchange Active.	Screenshot of live list and device status.
6. Modbus statistics	PLX Modbus Tx/Rx counts increase while RTAC polls.	Screenshot of Modbus statistics.
7. RTAC raw register check	Raw status and current/load registers update.	Screenshot of RTAC online tags.
8. RTAC bit check	Fault/warning/prewarning bits are decoded correctly.	Screenshot of Boolean tags.
9. HMI check	HMI values match RTAC tags.	Screenshot of HMI screen.

11. Troubleshooting Guide

Symptom	Most likely causes	First checks
PLX BF LED red	PROFIBUS cable issue, wrong termination, wrong baud, wrong address, GSD mismatch, bad slot config.	Check DB9 connectors, A/B, termination, PLX live list, event log.
SIMOCODE BUS LED off/blinking	No valid PROFIBUS data exchange.	Check PLX master online, node address, baud rate, cable, termination.
SIMOCODE GEN. FAULT red	SIMOCODE hardware/configuration fault, current module mismatch, missing system cable, active motor protection fault.	Go online in TIA/SIMOCODE ES and read diagnostics/error buffer.
PLX sees node but Data Exchange not active	Wrong GSD, wrong Basic Type/slot length, wrong station address, parameter mismatch.	Check PLX device status, identity, slot configuration, SIMOCODE project.
PLX PROFIBUS state forced Offline	Modbus/Logix communication fail behavior forcing the master offline.	Set Modbus Comms Fail to Force to Clear during bench/RTAC commissioning.
RTAC cannot read registers	Wrong IP, wrong Modbus interface, wrong register offset, PLX not acting as Modbus TCP server/slave, port blocked.	Ping PLX, check RTAC server IP, check PLX Modbus stats, verify Primary Interface.
RTAC values stay zero	No data exchange, wrong register offset, wrong poll quantity, real status bits are false.	Check PLX Data Exchange Active, read raw words, then check bit extraction.
HMI shows no fault but device is offline	HMI is only looking at fault bits, not communication status.	Add separate communication status tag and alarm.

11.1 Fast troubleshooting order

78. Do not change settings first. Take photos/screenshots of LEDs and status pages.
79. Check power to SIMOCODE, PLX, RTAC, and Ethernet switch.
80. Check SIMOCODE DEVICE/BUS/GEN. FAULT LEDs.
81. Check PLX OK/RUN/BF LEDs.
82. Check PROFIBUS cable and termination switch positions.
83. Open PLX live list and confirm each slave is Online and Data Exchange Active.
84. Open PLX event log and Modbus statistics.
85. Open RTAC tags and confirm raw registers update.
86. Only then check bit extraction and HMI graphics.

12. Adding Another SIMOCODE Later

Use this section when adding SIMOCODE 3, SIMOCODE 4, or any future device. The goal is to keep the project organized and avoid register overlap.

12.1 New device checklist

87. Choose the next unused PROFIBUS address. Example: SIMOCODE 3 uses node 4 if nodes 2 and 3 are already used.
88. Set the address in SIMOCODE using TIA/SIMOCODE ES or approved addressing method.
89. Check the current measuring module and system cable.
90. Set the correct motor protection Is1 current.
91. Check cyclic send data mapping.
92. Add the new device in PLX using the same SIMOCODE GSD.
93. Set the PLX device node address to match the SIMOCODE online address.
94. Allocate a new 10-register block. Example: SIMOCODE 3 base = 20.
95. Download PLX and confirm Data Exchange Active.
96. Add RTAC raw holding registers and bit extraction tags.
97. Add RTAC polls or expand the existing poll quantity.
98. Verify raw tags before adding HMI objects.
99. Add the HMI row or faceplate for the new motor/device.
100. Update Appendix A in this document.

12.2 Register pattern for future devices

Device number	PROFIBUS address	Register base	Status word	Imax % of Is	Control word
1	2	0	0	1	9
2	3	10	10	11	19
3	4	20	20	21	29
4	5	30	30	31	39
N	N + 1	(N - 1) x 10	Base + 0	Base + 1	Base + 9

Expansion rule: Keep at least 10 registers per device. This makes future devices easier to add and leaves room for extra values later.

Appendix A. Current Two-Device Address and Register Map

Device	PROFIBUS node	Signal	SIMOCODE source	PLX Modbus offset	RTAC tag	HMI tag
SIMOCODE 1	2	Overload prewarning	Status word bit 3	0 bit 3	SC1_OL_PREWARN	MTR01_OL_PREWARN
SIMOCODE 1	2	Group fault	Status word bit 6	0 bit 6	SC1_GROUP_FAULT	MTR01_FAULT
SIMOCODE 1	2	Group warning	Status word bit 7	0 bit 7	SC1_GROUP_WARNING	MTR01_WARNING
SIMOCODE 1	2	Imax % of Is	Cyclic send word	1	SC1_IMAX_PERCENT	MTR01_LOAD_PERCENT
SIMOCODE 2	3	Overload prewarning	Status word bit 3	10 bit 3	SC2_OL_PREWARN	MTR02_OL_PREWARN
SIMOCODE 2	3	Group fault	Status word bit 6	10 bit 6	SC2_GROUP_FAULT	MTR02_FAULT
SIMOCODE 2	3	Group warning	Status word bit 7	10 bit 7	SC2_GROUP_WARNING	MTR02_WARNING
SIMOCODE 2	3	Imax % of Is	Cyclic send word	11	SC2_IMAX_PERCENT	MTR02_LOAD_PERCENT

Appendix B. References and Open Items

B.1 Main references

- ProSoft PLX51-PBM User Manual, July 24, 2025.
- Siemens SIMOCODE pro System Manual, 02/2023.
- Siemens SIMOCODE pro for PROFIBUS System Manual, 08/2013.
- Siemens PROFIBUS Bus Connector Equipment Manual, 06/2022.
- SEL RTAC 3555 and ACSELERATOR RTAC manuals - verify final RTAC settings against the current site-approved manual.
- SEL-3555 RTAC product documentation page - use to confirm current instruction manuals, firmware, and product data before final release.
- SEL-5033 ACSELERATOR RTAC Software page - use to confirm current software version and RTAC project workflow.
- SEL-5035 ACSELERATOR Diagram Builder Software page - use to confirm HMI/Diagram Builder version compatibility and HMI project guidance.
- SEL RTAC Modbus Client/Server video series - optional training reference for adding and testing Modbus client sessions.

End of draft handover guide